Statics – Resolving Forces

A-Level Physics

For each question, draw a force diagram and resolve the forces horizontally and vertically. Assume all objects are in equilibrium.

Where required, take $g = 9.81\,\text{m}\,\text{s}^{-2}$.

1. A sign of weight 180 N is suspended by two cables. One cable makes an angle of 35° above the horizontal, and the other makes an angle of 50° above the horizontal.

   Calculate the tension in each cable.

2. A small ring is in equilibrium under three forces. One force is 80 N horizontally to the right. A second force is 120 N at 65° above the horizontal to the left.

   Calculate the **magnitude and direction** of the third force.

3. A 25 kg stage light is supported by two cables. The left cable makes an angle of 40° above the horizontal, while the right cable makes an angle of 25° above the horizontal.

   Calculate the tension in each cable.

4. A 30 kg box rests on a horizontal floor. A force of 100 N is applied at 30° below the horizontal. The box remains stationary.

   Calculate:

   (a) the normal reaction force;

   (b) the frictional force.

5. A ring is acted upon by three forces. A 90 N force acts horizontally to the left and a 140 N force acts at 55° above the horizontal to the right.

   The third force keeps the ring in equilibrium.

   Calculate the **magnitude and direction** of the third force.

6. A 300 N object is held stationary by two ropes. One rope acts at 20° above the horizontal to the left, and the other at 60° above the horizontal to the right.

   Calculate the tension in each rope.

Statics – Answers

1. 35° cable: 116 N
   50° cable: 148 N

2. Third force magnitude: 113 N
   Direction: 74.9° below the horizontal to the left

3. Left cable tension: 245 N
   Right cable tension: 207 N

4. Normal reaction: 344 N
   Frictional force: 86.6 N

5. Third force magnitude: 115 N
   Direction: 85.2° below the horizontal to the right

6. 20° rope tension: 152 N
   60° rope tension: 286 N